

Mohammad Taha Askari

PHD CANDIDATE · ELECTRICAL AND COMPUTER ENGINEERING, THE UNIVERSITY OF BRITISH COLUMBIA

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Summary

PhD candidate in Electrical and Computer Engineering at the University of British Columbia, researching machine learning, probabilistic modeling, and sequence models. My work focuses on neural sequence models that learn joint probability distributions for systems with memory and uncertainty, applied to communication systems with nonlinear dynamics. I have extensive experience conducting independent research, publishing in peer-reviewed venues, and building end-to-end ML pipelines in Python that connect theory, experimentation, and scalable implementation.

Research Experience

M.A.Sc. & PhD Thesis at Data Communications Group

ECE Department, UBC

RESEARCH ASSISTANT (SUPERVISOR: PROF. LUTZ LAMPE)

Sep. 2020 - Sep. 2026

- Conducted independent research on **probabilistic modeling and sequence design** for systems with long-range dependencies and nonlinear dynamics.
- Developed an **analytical filter-based model** to characterize how learned distributions interact with a nonlinear channel.
- Proposed a **rejection sampling / sequence selection algorithm** that improves robustness by exploiting temporal dependencies in generated sequences.
- Designed a **Bayesian inference algorithm** incorporating prior distributions for phase estimation in the low-SNR regime.
- Led research on **joint-distribution learning** for nonlinear channels with memory, progressing from autoregressive sequence-based models to Transformer-based sequential architectures.

Research Internship at Nokia Bell Labs

Greater Paris Metropolitan Region, France

MACHINE LEARNING RESEARCH INTERN

April. 2025 - July. 2026

- Designed and implemented an **autoregressive sequence modeling framework** using neural sequence models and **Gumbel-Softmax** sampling for joint distribution learning.
- Built an **end-to-end differentiable training pipeline** integrating neural models with a differentiable physical simulator, enabling gradient-based optimization over generated sequences.
- Extended the framework to a **constrained output space**, achieving deployable gains over classical combinatorial and sampling-based baselines.
- Developed a **Transformer-based sequential architecture**; refined the training objective with a **rate-aware penalty** to preserve achievable rate gains, outperforming classical and sampling-based baselines.
- Evaluated proposed models in **multi-dimensional settings**, demonstrating improved robustness and scalability over heuristic and sampling-based methods.

Research Internship at Roche Canada

Greater Toronto Area, Canada

ALGORITHM R&D SOFTWARE ENGINEERING INTERN

Jun. 2024 - Oct. 2024

- Designed and implemented **ML pipelines** for feature extraction, dataset generation, and neural network training for large-scale sequencing data.
- Developed and optimized **weakly supervised deep learning models**, including a custom **multi-label architecture** for multi-class anomaly detection and segmentation.
- Integrated **model interpretability, performance analysis, and computational complexity evaluation** into the training workflow to support iterative model improvement.

Publications and Posters

- M.T. Askari, L. Lampe, and A. Ghazisaeidi "Sequential Neural Probabilistic Amplitude Shaping: Learning the Channel's Language," 2026 European Conference on Optical Communication (ECOC).
- M.T. Askari, L. Lampe, and A. Ghazisaeidi "Neural Probabilistic Amplitude Shaping for Nonlinear Fiber Channels," 2026 Optical Fiber Communications Conference and Exhibition (OFC). [Online Access]
- M.T. Askari, L. Lampe, and A. Ghazisaeidi "Neural Probabilistic Shaping: Joint Distribution Learning for Optical Fiber Communications," 2025 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari and L. Lampe, "Probabilistic Shaping for Nonlinearity Tolerance," Journal of Lightwave Technology. [Online Access]

- M.T. Askari and L. Lampe, "Perturbation-based Sequence Selection for Probabilistic Amplitude Shaping," 2024 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari and L. Lampe, "Bayesian Phase Search for Probabilistic Amplitude Shaping," 2023 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari, "Interplay between Fiber Nonlinearity and Probabilistic Amplitude Shaping," Master's Thesis, UBC [Online Access]
- M.T. Askari, L. Lampe, and J. Mitra, "Probabilistic Amplitude Shaping and Nonlinearity Tolerance: Analysis and Sequence Selection Method," Journal of Lightwave Technology. [Online Access]
- M.T. Askari, L. Lampe, and J. Mitra, "Nonlinearity Tolerant Shaping with Sequence Selection," 2022 European Conference on Optical Communication (ECOC) [Online Access]
- M.T. Askari, "Nonlinearity Tolerant Sequence Selection," Poster at 17th Canadian Workshop on Information Theory (CWIT)

Research Interests

Machine Learning for Sequential and Structured Data
Probabilistic Modeling and Joint Distribution Learning
Neural Sequence Models (Autoregressive Models, Transformers)
End-to-End Learning and Optimization
Information Theory and Statistical Inference

Education

PhD in Electrical & Computer Engineering

Vancouver, Canada

ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT, THE UNIVERSITY OF BRITISH COLUMBIA (UBC)

Sep. 2022 - Sep. 2026

- Advisor: Prof. Lutz Lampe

M.A.Sc. in Electrical & Computer Engineering

Vancouver, Canada

ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT, THE UNIVERSITY OF BRITISH COLUMBIA (UBC)

Sep. 2020 - Aug. 2022

- Cumulative GPA: 93.7 out of 100 (4.0/4.0) via 18 credits
- Advisor: Prof. Lutz Lampe

B.Sc. in Electrical Engineering

Tehran, Iran

DEPARTMENT OF ELECTRICAL ENGINEERING, SHARIF UNIVERSITY OF TECHNOLOGY (SUT)

Sep. 2015 - Jul. 2020

- Cumulative GPA: 18.42 out of 20 (4.0/4.0) via 146 credits
- Advisor: Prof. Hamid Behroozi

Software Proficiency

Programming Python, Matlab, C/C++, Git, Julia, R

Relevant Courses

EECE 571F-Advanced Topics in Deep Learning	87.0 out of 100	Dr. Liao	UBC	Winter2 2025
CPSC 436N-Natural Language Processing	95.0 out of 100	Dr. Shwartz	UBC	Winter1 2023
EECE 571D-Detection, Estimation, and Learning	98.0 out of 100	Dr. Lampe	UBC	Winter2 2021
EECE 562-Statistical Signal Processing	90.0 out of 100	Dr. Wang	UBC	Winter1 2021
CPSC 540-Advanced Machine Learning	96.0 out of 100	Dr. Schmidt	UBC	Winter2 2020
CPSC 340-Machine Learning & Data Mining	100 out of 100	Dr. Wood	UBC	Winter1 2020
EECE 565-Communication Networks	95.0 out of 100	Dr. Wong	UBC	Winter1 2020
Introduction to Machine Learning	18.0 out of 20	Dr. Mohammadzade	SUT	Spring 2019
Information & Coding Theory	18.7 out of 20	Dr. Mirmohseni	SUT	Fall 2018

Selected Course Projects

COVID-19 Detection: A Combination of Transformer Models with Image Augmentation Methods

Winter 2020

COURSE: ADVANCED MACHINE LEARNING

UBC

- Studied methods for generating synthetic X-ray images to improve the accuracy of COVID-19 detection models and implemented GAN and VAE to augment data.
- Applied transformer models to COVID-19 classification based on chest X-ray images.
- Supervisor: Dr. Schmidt